

Derivatives and Fixed Income

PGE M1 Fall 2026

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Session 6: The Yield Curve & Swaps



About me

- ▷ Office: B3.09
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Housekeeping



Scheduling

- ▷ 2 sessions on Tuesday Oct. 6
- ▷ Bring your laptop for Tuesday's session!!

T/F



If two bonds have the same maturity, the bond with the higher coupon rate has the lower duration.



Let Bond *A* have a duration of 9.3 and a convexity of 23. Suppose yields increase by 2 percentage points.

- a. The effect on the change in prices due to convexity alone is a price decrease of .92%.



Suppose you purchase a 1-year zero coupon bond with a face value of \$1 at .9091 today. Assume the 1-year spot rate is 10%.

- a. The expected return of holding this bond until maturity is 10%.



Suppose you purchase a 1-year zero coupon bond with a face value of \$1 at .9091 today. Assume the 1-year spot rate is 10%.

- b. The yield-to-maturity of this bond is 10%.



Suppose you purchase a 1-year zero coupon bond with a face value of \$1 at .9091 today. Assume the 1-year spot rate is 10%.

- c. There is another bond in the market with a maturity of 1 year, paying a semi-annual coupon rate of 8%. Assume this bond is fairly valued, the yield curve is currently upward sloping, and both bonds are issued by the same issuer. T/F: The yield-to-maturity on this bond is also 10%.

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- a. One year later, the new 1-year spot rate is 6%. Assuming no arbitrage holds, what is the new price of bond *B*?

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- b. If you sold bond *B* at the new market price, what would be the percentage return on this investment? Compare this to the return on the 1-year ZCB.

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- c. For comparison, multiply the original price of the 2-year ZCB by the 1-year spot rate from 1-year ago (4%).

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- d. Given the new 1-year spot rate, for the potential buyer of bond *B*, what would be the expected return of holding this bond until maturity?

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- e. If you did not sell bond *B* but held it until maturity (over the 2 full years), what would be the average rate of return on your investment?

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- f. Suppose, instead of the new spot rate being 6%, it was actually 7%. Compute the new price of bond *B*, and the expected return of selling the bond after 1-year.

Building the Yield Curve



Suppose the 1-year and 2-year spot rates are currently 4% and 5%. Bond *A* is a 1-year ZCB with price 96.15 and bond *B* is a 2-year ZCB with a price 90.70. Both bonds are issued by the same issuer and have a face value of 100.

- g. For new investors, is the new 1-year spot rate of 7% a good deal?

Building the Yield Curve



For a 2-year ZCB, we can either “lock-in” the 2-year discount rate today, or “roll-over” the 1-year spot rates.

▷ “Locking-in”

$$PV_0 = \frac{C}{(1 + s_0^{(2)})^2}$$

▷ “Rolling-over”

$$PV_0 = \frac{C}{(1 + s_0^{(1)})(1 + \mathbb{E}_0(1 + s_1^{(1)}))}$$

▷ This implies

$$1 + s_0^{(2)} \approx \sqrt{(1 + s_0^{(1)})(1 + \mathbb{E}_0(1 + s_1^{(1)}))} + \text{risk premium}$$



Implications of the Yield Curve

Starting from short-term spot rates, we can repeat this procedure to build the expected value of the new 1-year spot rate, N years from today. The shape of the yield curve, therefore, reveals expectations about how rates will move in the future.



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Forward Rates



Going back to

$$1 + s_0^{(2)} \approx \sqrt{(1 + s_0^{(1)})(1 + \mathbb{E}_0(1 + s_1^{(1)}))} + \text{risk premium}$$

We call $\mathbb{E}_0(1 + s_1^{(1)})$ the **forward rate** between years 1 and 2, denoted $F_{1,2}$.

Ignoring the risk premium, we can compute the forward rate between any two years T_1 and T_2 as

$$(1 + F_{T_1, T_2})^{T_2 - T_1} = \frac{(1 + s_0^{(T_2)})^{T_2}}{(1 + s_0^{(T_1)})^{T_1}}$$

Forward Rates



Example. Let the 1-year spot rate be 4% and the 2-year spot rate be 5%. Then,

$$(1 + F_{T_1, T_2})^{T_2 - T_1} = \frac{(1 + s_0^{(T_2)})^{T_2}}{(1 + s_0^{(T_1)})^{T_1}}$$

$$(1 + F_{1,2})^1 = \frac{(1 + .05)^2}{(1 + .04)^1}$$

$$F_{1,2} \approx 6\%$$



Why does the central bank change rates?

In a growing economy, the central bank raises rates to promote savings and price stability.

- ▷ Upward sloping YC implies investors expect the economy to grow in the future
- ▷ Cost of borrowing for corporations increase

Good economy = bad for tech stocks



“Friday’s rout began after a much stronger than expected US jobs report triggered bets that the Fed will raise rates this year ... Higher yields are caustic to fast-growing, highly valued tech companies because they reduce the allure of their future cash flows.”

Nasdaq tumbles 4% as shares in chip and memory groups sink,
Financial Times, 2026.

Misc. Topics: Par Yield



Suppose the average cost of borrowing for a corporation over a maturity of 5 years is 10%. If the bond issues a 5-year coupon bond, what coupon rate should it set?

Misc. Topics: Par Yield



The **par yield** is the coupon rate at which a newly issued coupon bond, given the underlying spot rates, trades at par.

- ▷ Coupons are “like” interest payments on debt, set by the issuer to attract investors
- ▷ If coupon rate $>$ cost of borrowing, bad deal for issuer
- ▷ If coupon rate $<$ cost of borrowing, demand for bonds may be under-subscribed

T/F



A 3-year coupon bond was initially issued at par. One year, it is trading at a premium. What must be true?

Misc. Topics: Par Yield



Let CR be the coupon rate for a N -period bond. Let $DF^{(n)}$ be the n -maturity discount factor. For example, if the 3-year spot rate is 5%,

$$DF^{(3)} = \frac{1}{1.05^3} = .8638$$

The par yield is given by

$$CR = \frac{1 - DF^{(N)}}{\sum_{n=1}^N DF^{(n)}}$$

Misc. Topics: Par Yield



Example. Suppose spot rates are currently: $s^{(1)} = 3\%$, $s^{(2)} = 3.5\%$, $s^{(3)} = 4\%$. Then,

$$DF^{(1)} = \frac{1}{1.03} = 0.9709$$

$$DF^{(2)} = \frac{1}{1.035^2} = 0.9335$$

$$DF^{(3)} = \frac{1}{1.04^3} = 0.8890$$

so $\sum_{n=1}^N DF^{(n)} = 0.9709 + 0.9335 + 0.8890 = 2.7934$. Then, the par yield for a 3-year coupon bond is

$$CR = \frac{1 - DF^{(N)}}{\sum_{n=1}^N DF^{(n)}} = \frac{1 - 0.8890}{2.7934} = 3.97\%$$

Misc. Topics: Credit Rating Scale



	Moody's	S&P / Fitch	Risk-free +
Investment Grade	Aaa	AAA	0.3%–0.6%
	Aa1–Aa3	AA+/AA/AA-	0.5%–0.9%
	A1–A3	A+/A/A-	0.7%–1.3%
	Baa1–Baa3	BBB+/BBB/BBB-	1.2%–2.2%
Speculative Grade	Ba1–Ba3	BB+/BB/BB-	2.5%–4.5%
	B1–B3	B+/B/B-	4.5%–7%
	Caa1–Caa3	CCC+/CCC/CCC-	7%–12%+
	Ca	CC/C	12%+ (distressed)
	C/D	D	N/A

Interest Rate Swaps



Given a notional principal, an **interest rate swap** is a agreement between two parties in which, over the life of the contract, party *A* pays party *B* interest at a fixed rate, while party *B* pays party *A* interest at a floating (variable) rate.

The Scenario



Air France

**Borrowed floating.
Hates surprises.**

Airlines carry heavy floating-rate debt for fleets and leases. Rate hikes squeezes their budget.

Wants: predictable, fixed payments

Microsoft

**Locked into a 30-year fixed.
Doesn't need the safety.**

Borrows cheaply in public bond markets, but coupon rates stay fixed.

Wants: floating exposure, lower long-run cost

Mechanics of a Swap



	Fixed rate	Floating rate
Microsoft	4.0%	SOFR – 0.3%
Air France–KLM	6.5%	SOFR + 1.0%

We assume here, that the secured overnight financing rate (SOFR) is the reference rate for this transaction.

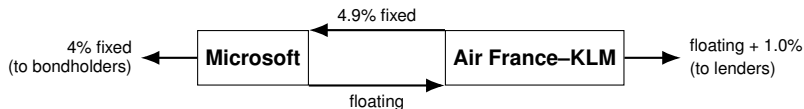
Mechanics of a Swap



	Fixed rate	Floating rate
Microsoft	4.0%	SOFR – 0.3%
Air France–KLM	6.5%	SOFR + 1.0%
Microsoft's advantage	2.5%	1.3%

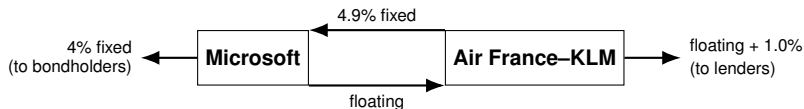
Even though MSFT has lower borrowing in **both** fixed and floating rates, we say that Microsoft has a comparative advantage in fixed rates while Air France has a comparative advantage in floating rates.
(Air France is less bad in floating rates than it is in fixed rates)

Mechanics of a Swap



In the contract, suppose Microsoft pays Air France a floating rate of interest on the agreed principal while Air France pays Microsoft a fixed rate of 4.9% every year.

Mechanics of a Swap

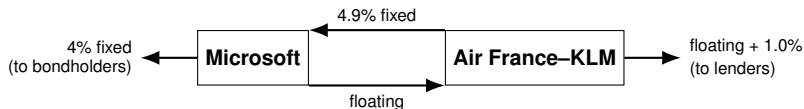


Microsoft

- 4% to bondholders
- floating to Air France
- + 4.9% from Air France

$$\begin{aligned} & - \text{floating} + 0.9\% \\ & = \text{pay floating} - 0.9\% \end{aligned}$$

Mechanics of a Swap



Air France

– (floating + 1.0%) to lenders

+ floating from Microsoft

– 4.9% to Microsoft

– 5.9%
= pay 5.9% fixed

Mechanics of a Swap



	Fixed rate	Floating rate
Microsoft	4.0%	SOFR – 0.3%
Air France–KLM	6.5%	SOFR + 1.0%
Microsoft's advantage	2.5%	1.3%

After the swap, Microsoft pays $\text{SOFR} - 0.9\%$ while Air France pays a fixed rate of 5.9%.

Mechanics of a Swap



	Direct	Via Swap	Net Gain
Microsoft	SOFR – 0.3%	SOFR – 0.9%	0.6%
Air France–KLM	6.5%	5.9%	0.6%
			1.2%

The total gain is determined by the **quality spread differential**, the difference in comparative advantage in fixed minus the comparative advantage in floating.

(What matters is that the difference is not zero.)

T/F

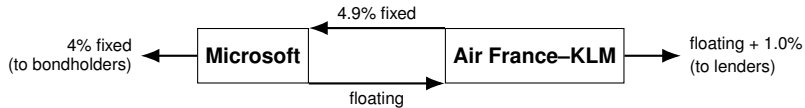


An interest rate swap can benefit both parties even if one party has an absolute advantage in both fixed and floating rates.

Designing a Swap



How do we know Air France should pay Microsoft 4.9% fixed?



Designing a Swap



	Fixed rate	Floating rate
Microsoft	4.0%	SOFR – 0.3%
Air France–KLM	6.5%	SOFR + 1.0%
Microsoft's advantage	2.5%	1.3%

Since the QSD is equal to 1.2%, evenly splitting the benefits implies Air France, which wants to borrow fixed, will eventually pay $6.5\% - 0.6\% = 5.9\%$. It pays Microsoft 5.9% minus its spread over the SOFR, so $5.9\% - 1\% = 4.9\%$.

Exercise



	Fixed rate	Floating rate
Company A	5.0%	SOFR + 0.1%
Company B	6.4%	SOFR + 0.5%

Company A requires a floating-rate loan while Company B requires a fixed-rate loan. Design a swap in which the parties equally share the benefits.

Exercise



	Fixed rate	Floating rate
Company A	5.0%	SOFR + 0.1%
Company B	6.4%	SOFR + 0.5%

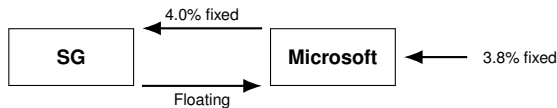
Exercise 7.1



	Fixed rate	Floating rate
Company A	5.0%	SOFR + 0.1%
Company B	6.4%	SOFR + 0.6%

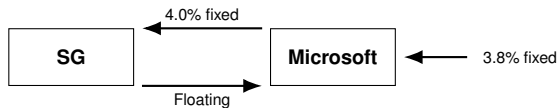
Company A requires a floating-rate loan while Company B requires a fixed-rate loan. Design a swap that will net a bank, acting as an intermediary, 0.1% per annum, in which both companies equally share the benefits.

Other Uses of Swaps



Suppose Microsoft purchased government bonds and receives a fixed coupon rate of 3.8% every year. It wants exposure to a possible interest rate hike and enters into a swap agreement with Société Générale.

Other Uses of Swaps



Microsoft

+3.8% from US govt

−4% to SG

+floating from SG

+ floating − 0.20%
= receive floating − 0.20%